

RKDF University

Program Outcomes, Program Specific Outcomes and Course Outcomes

Faculty of Pharmacy

Bachelor of Pharmacy (B. Pharm)

Programme Outcome (POs)

- PO1- Demonstrate knowledge of the basic pharmaceutical sciences and the ability to acquire, manage and use current information for problem solving.
- PO2- Describe the synthesis, formulation, analysis and pharmacological aspects of drugs and pharmaceuticals.
- PO3- Identify the rules and regulations involved in the drug discovery and development, manufacture, distribution and sale of medicines.
- PO4- Observe record, analyze, criticize, organize, improvise and manage documents, data and information related to pharmaceutical products and practices.
- PO5- Develop problem-based learning approach and analytical thinking in his/her academic and professional life.
- PO6- Demonstrate the ability to plan and implement professional activities.
- PO7- Act efficiently as a leader in the diverse areas of the profession.
- PO8- Write, interpret and communicate effectively and scientifically.
- PO9- Apply the knowledge and skills gained through education to gain recognition in professional circle and society.
- PO10- Partnering with other health care communities to provide innovative solutions.
- PO11- Create awareness in society about the effective and safe use of medicines.
- PO12- Demonstrate eco-friendly products and processes to maintain public health.

PO13- Imbibe ethical practices and moral values in personal and professional endeavors.

PO14- Tackle future challenges through lifelong learning.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1- The students shall be able to apply the knowledge and skill gained from various subjects and the ability developed throughout the course of the program in performing a job either independently or as a member of a team in various fields of pharmacy profession.

PSO2- The students shall be able to make an initiation and achieve an innovation by properly integrating the input from various resources either independently or as a member of a team in various fields of pharmacy profession for betterment of the quality of life of the patients in the society

COURSE OUTCOMES (COs)

BP101T. HUMAN ANATOMY AND PHYSIOLOGY-I (Theory)

CO1: Explain the gross morphology, structure and functions of various organs of the human body.

CO2: Describe the various homeostatic mechanisms and their imbalances.

CO3: Identify the various tissues and organs of different systems of human body.

CO4: Perform the various experiments related to special senses and nervous system.

CO5: Appreciate coordinated working pattern of different organs of each system

BP107P. HUMAN ANATOMY AND PHYSIOLOGY (Practical)

CO1: Practical physiology is complimentary to the theoretical discussions in physiology.

CO2: Practicals allow the verification of physiological processes discussed in theory classes through experiments on living tissue, intact animals or normal human beings.

BP102T. PHARMACEUTICAL ANALYSIS (Theory)

CO1: Understand the principles of volumetric and electro chemical analysis

CO2: Carryout various volumetric and electrochemical titrations

CO3: Develop analytical skills

BP103T. PHARMACEUTICS- I (Theory)

CO1: Know the history of profession of pharmacy

CO2: Understand the basics of different dosage forms, pharmaceutical incompatibilities and pharmaceutical calculations

CO3: Understand the professional way of handling the prescription

CO4: Preparation of various conventional dosage forms

BP104T. PHARMACEUTICAL INORGANIC CHEMISTRY (Theory)

CO1: know the sources of impurities and methods to determine the impurities in inorganic drugs and pharmaceuticals

CO2: understand the medicinal and pharmaceutical importance of inorganic compounds

BP105T.COMMUNICATION SKILLS (Theory)

CO1: Understand the behavioral needs for a Pharmacist to function effectively in the areas of pharmaceutical operation

CO2: Communicate effectively (Verbal and Non Verbal)

CO3: Effectively manage the team as a team player

CO4: Develop interview skills

CO5: Develop Leadership qualities and essentials

BP 106RBT.REMEDIAL BIOLOGY (Theory)

CO1: know the classification and salient features of five kingdoms of life

CO2: understand the basic components of anatomy & physiology of plant

CO3: know understand the basic components of anatomy & physiology animal with special reference to human

BP 106RMT.REMEDIAL MATHEMATICS (Theory)

CO1: Know the theory and their application in Pharmacy

CO2: Solve the different types of problems by applying theory

CO3: Appreciate the important application of mathematics in Pharmacy

BP 201 T. HUMAN ANATOMY AND PHYSIOLOGY-II (Theory)

CO1: Explain the gross morphology, structure and functions of various organs of the human body.

CO2: Describe the various homeostatic mechanisms and their imbalances.

- CO3: Identify the various tissues and organs of different systems of human body.
- CO4: Perform the hematological tests like blood cell counts, haemoglobin estimation, bleeding/clotting time etc and also record blood pressure, heart rate, pulse and respiratory volume.
- CO5: Appreciate coordinated working pattern of different organs of each system
- CO6: Appreciate the interlinked mechanisms in the maintenance of normal functioning (homeostasis) of human body.

BP202T. PHARMACEUTICAL ORGANIC CHEMISTRY –I (Theory)

- CO1: Write the structure, name and the type of isomerism of the organic compound
- CO2: Write the reaction, name the reaction and orientation of reactions
- CO3: Account for reactivity/stability of compounds,
- CO4: Identify/confirm the identification of organic compound

BP203 T. BIOCHEMISTRY (Theory)

- CO1: Understand the catalytic role of enzymes, importance of enzyme inhibitors in design of new drugs, therapeutic and diagnostic applications of enzymes.
- CO2: Understand the metabolism of nutrient molecules in physiological and pathological conditions.
- CO3: Understand the genetic organization of mammalian genome and functions of DNA in the synthesis of RNAs and proteins.

BP 204T.PATHOPHYSIOLOGY (THEORY)

- CO1: Describe the etiology and pathogenesis of the selected disease states;
- CO2: Name the signs and symptoms of the diseases; and
- CO3: Mention the complications of the diseases.

BP205 T. COMPUTER APPLICATIONS IN PHARMACY (Theory)

- CO1: know the various types of application of computers in pharmacy
- CO2: know the various types of databases
- CO3: know the various applications of databases in pharmacy

BP 206 T. ENVIRONMENTAL SCIENCES (Theory).

- CO1: Create the awareness about environmental problems among learners.
- CO2: Impart basic knowledge about the environment and its allied problems.
- CO3: Develop an attitude of concern for the environment.
- CO4: Motivate learner to participate in environment protection and environment improvement.

- CO5: Acquire skills to help the concerned individuals in identifying and solving environmental problems.
- CO6: Strive to attain harmony with Nature.

BP301T. PHARMACEUTICAL ORGANIC CHEMISTRY –II (Theory)

- CO1: Write the structure, name and the type of isomerism of the organic compound
- CO2: Write the reaction, name the reaction and orientation of reactions
- CO3: Account for reactivity/stability of compounds,
- CO4: Prepare organic compounds

BP302T. PHYSICAL PHARMACEUTICS-I (Theory)

- CO1: Understand various physicochemical properties of drug molecules in the designing the dosage forms
- CO2: Know the principles of chemical kinetics & to use them for stability testing and determination of expiry date of formulations
- CO3: Demonstrate use of physicochemical properties in the formulation development and evaluation of dosage forms.

BP 303 T. PHARMACEUTICAL MICROBIOLOGY (Theory)

- CO1: Understand methods of identification, cultivation and preservation of various microorganisms
- CO2: To understand the importance and implementation of sterilization in pharmaceutical processing and industry
- CO3: Learn sterility testing of pharmaceutical products.
- CO4: Carried out microbiological standardization of Pharmaceuticals.
- CO5: Understand the cell culture technology and its applications in pharmaceutical industries.

BP401T. PHARMACEUTICAL ORGANIC CHEMISTRY –III (Theory)

- CO1: understand the methods of preparation and properties of organic compounds
- CO2: explain the stereo chemical aspects of organic compounds and stereo chemical reactions
- CO3: know the medicinal uses and other applications of organic compounds

BP402T. MEDICINAL CHEMISTRY – I (Theory)

- CO1: understand the chemistry of drugs with respect to their pharmacological activity
- CO2: understand the drug metabolic pathways, adverse effect and therapeutic value of drugs

CO3: know the Structural Activity Relationship (SAR) of different class of drugs

CO4: write the chemical synthesis of some drugs

BP 403 T. PHYSICAL PHARMACEUTICS-II (Theory)

CO1: Understand various physicochemical properties of drug molecules in the designing the dosage forms

CO2: Know the principles of chemical kinetics & to use them for stability testing and determination of expiry date of formulations

CO3: Demonstrate use of physicochemical properties in the formulation development and evaluation of dosage forms.

BP 404 T. PHARMACOLOGY-I (Theory)

CO1: Understand the pharmacological actions of different categories of drugs

CO2: Explain the mechanism of drug action at organ system/sub cellular/ macromolecular levels.

CO3: Apply the basic pharmacological knowledge in the prevention and treatment of various diseases.

CO4: Observe the effect of drugs on animals by simulated experiments

CO5: Appreciate correlation of pharmacology with other bio medical sciences

BP 405 T. PHARMACOGNOSY AND PHYTOCHEMISTRY I (Theory)

CO1: to know the techniques in the cultivation and production of crude drugs

CO2: to know the crude drugs, their uses and chemical nature

CO3: know the evaluation techniques for the herbal drugs

CO4: to carry out the microscopic and morphological evaluation of crude drugs

BP501T. MEDICINAL CHEMISTRY – II (Theory)

CO1: Understand the chemistry of drugs with respect to their pharmacological activity

CO2: Understand the drug metabolic pathways, adverse effect and therapeutic value of drugs

CO3: Know the Structural Activity Relationship of different class of drugs

CO4: Study the chemical synthesis of selected drugs

BP 502 T. Industrial Pharmacy I (Theory)

CO1: Know the various pharmaceutical dosage forms and their manufacturing techniques.

CO2: Know various considerations in development of pharmaceutical dosage forms

CO3: Formulate solid, liquid and semisolid dosage forms and evaluate them for their quality

BP503.T. PHARMACOLOGY-II (Theory)

- CO1: Understand the mechanism of drug action and its relevance in the treatment of different diseases
- CO2: Demonstrate isolation of different organs/tissues from the laboratory animals by simulated experiments
- CO3: Demonstrate the various receptor actions using isolated tissue preparation
- CO4: Appreciate correlation of pharmacology with related medical sciences

BP504 T. PHARMACOGNOSY AND PHYTOCHEMISTRY II (Theory)

- CO1: to know the modern extraction techniques, characterization and identification of the herbal drugs and phytoconstituents
- CO2: to understand the preparation and development of herbal formulation.
- CO3: to understand the herbal drug interactions
- CO4: to carryout isolation and identification of phytoconstituents

BP 505 T. PHARMACEUTICAL JURISPRUDENCE (Theory)

- CO1: The Pharmaceutical legislations and their implications in the development and marketing of pharmaceuticals.
- CO2: Various Indian pharmaceutical Acts and Laws
- CO3: The regulatory authorities and agencies governing the manufacture and sale of pharmaceuticals
- CO4: The code of ethics during the pharmaceutical practice

BP601T. MEDICINAL CHEMISTRY – III (Theory)

- CO1: Understand the importance of drug design and different techniques of drug design.
- CO2: Understand the chemistry of drugs with respect to their biological activity.
- CO3: Know the metabolism, adverse effects and therapeutic value of drugs.
- CO4: Know the importance of SAR of drugs.

BP602 T. PHARMACOLOGY-III (Theory)

- CO1: Understand the mechanism of drug action and its relevance in the treatment of different infectious diseases
- CO2: Comprehend the principles of toxicology and treatment of various poisonings and
- CO3: Appreciate correlation of pharmacology with related medical sciences.

BP 603 T. HERBAL DRUG TECHNOLOGY (Theory)

- CO1: understand raw material as source of herbal drugs from cultivation to herbal drug product

- CO2: know the WHO and ICH guidelines for evaluation of herbal drugs
- CO3: know the herbal cosmetics, natural sweeteners, nutraceuticals
- CO4: Appreciate patenting of herbal drugs, GMP .

BP 604 T. Biopharmaceutics and Pharmacokinetics (Theory)

- CO1: Understand the basic concepts in biopharmaceutics and pharmacokinetics and their significance.
- CO2: Use of plasma drug concentration-time data to calculate the pharmacokinetic parameters to describe the kinetics of drug absorption, distribution, metabolism, excretion, elimination.
- CO3: To understand the concepts of bioavailability and bioequivalence of drug products and their significance.
- CO4: Understand various pharmacokinetic parameters, their significance & applications.

BP 605 T. PHARMACEUTICAL BIOTECHNOLOGY (Theory)

- CO1: Understanding the importance of Immobilized enzymes in Pharmaceutical Industries
- CO2: Genetic engineering applications in relation to production of pharmaceuticals
- CO3: Importance of Monoclonal antibodies in Industries
- CO4: Appreciate the use of microorganisms in fermentation technology

BP606TPHARMACEUTICAL QUALITY ASSURANCE (Theory)

- CO1: Understand the cGMP aspects in a pharmaceutical industry
- CO2: Appreciate the importance of documentation
- CO3: Understand the scope of quality certifications applicable to pharmaceutical industries
- CO4: Understand the responsibilities of QA & QC departments

BP701T. INSTRUMENTAL METHODS OF ANALYSIS (Theory)

- CO1: Understand the interaction of matter with electromagnetic radiations and its applications in drug analysis
- CO2: Understand the chromatographic separation and analysis of drugs.
- CO3: Perform quantitative & qualitative analysis of drugs using various analytical instruments.

BP 702 T. INDUSTRIAL PHARMACYII (Theory)

- CO1: Know the process of pilot plant and scale up of pharmaceutical dosage forms
- CO2: Understand the process of technology transfer from lab scale to commercial batch
- CO3: Know different Laws and Acts that regulate pharmaceutical industry

CO4: Understand the approval process and regulatory requirements for drug products

BP 703 T. PHARMACY PRACTICE (Theory)

CO1: Know various drug distribution methods in a hospital

CO2: Appreciate the pharmacy stores management and inventory control

CO3: Monitor drug therapy of patient through medication chart review and clinical review

CO4: Obtain medication history interview and counsel the patients

CO5: Identify drug related problems

CO6: Detect and assess adverse drug reactions

CO7: Interpret selected laboratory results (as monitoring parameters in therapeutics) of specific disease states

CO8: Know pharmaceutical care services

CO9: Do patient counseling in community pharmacy;

CO10: Appreciate the concept of rational drug therapy.

BP 704T: NOVEL DRUG DELIVERY SYSTEMS (Theory)

CO1: To understand various approaches for development of novel drug delivery systems.

CO2: To understand the criteria for selection of drugs and polymers for the development of Novel drug delivery systems, their formulation and evaluation

BP801T. BIOSTATISTICS AND RESEARCH METHODOLOGY (Theory)

CO1: Know the operation of M.S. Excel, SPSS, R and MINITAB[®], DoE (Design of Experiment)

CO2: Know the various statistical techniques to solve statistical problems

CO3: Appreciate statistical techniques in solving the problems.

BP 802T SOCIAL AND PREVENTIVE PHARMACY

CO1: Acquire high consciousness/realization of current issues related to health and pharmaceutical problems within the country and worldwide.

CO2: Have a critical way of thinking based on current healthcare development.

CO3: Evaluate alternative ways of solving problems related to health and pharmaceutical issues

BP803ET. PHARMA MARKETING MANAGEMENT (Theory)

CO1: After the successful completion of this course, the student shall be able to understanding of marketing concepts and techniques and their applications in the pharmaceutical industry.

BP804 ET: PHARMACEUTICAL REGULATORY SCIENCE (Theory)

- CO1: Know about the process of drug discovery and development
- CO2: Know the regulatory authorities and agencies governing the manufacture and sale of pharmaceuticals
- CO3: Know the regulatory approval process and their registration in Indian and international markets

BP 805T: PHARMACOVIGILANCE (Theory)

- CO1: Why drug safety monitoring is important?
- CO2: History and development of pharmacovigilance
- CO3: National and international scenario of pharmacovigilance
- CO4: Dictionaries, coding and terminologies used in pharmacovigilance
- CO5: Detection of new adverse drug reactions and their assessment
- CO6: International standards for classification of diseases and drugs
- CO7: Adverse drug reaction reporting systems and communication in pharmacovigilance
- CO8: Methods to generate safety data during pre clinical, clinical and post approval phases of drugs' life cycle
- CO9: Drug safety evaluation in paediatrics, geriatrics, pregnancy and lactation
- CO10: Pharmacovigilance Program of India (PvPI) requirement for ADR reporting in India
- CO11: ICH guidelines for ICSR, PSUR, expedited reporting, pharmacovigilance planning
- CO12: CIOMS requirements for ADR reporting
- CO13: Writing case narratives of adverse events and their quality.

BP 806 ET. Quality Control and Standardization Of Herbals (Theory)

- CO1: know WHO guidelines for quality control of herbal drugs
- CO2: know Quality assurance in herbal drug industry
- CO3: know the regulatory approval process and their registration in Indian and international markets
- CO4: appreciate EU and ICH guidelines for quality control of herbal drugs

BP 807 ET. COMPUTER AIDED DRUG DESIGN (Theory)

- CO1: Design and discovery of lead molecules
- CO2: The role of drug design in drug discovery process
- CO3: The concept of QSAR and docking
- CO4: Various strategies to develop new drug like molecules.
- CO5: The design of new drug molecules using molecular modeling software

BP808ET: CELL AND MOLECULAR BIOLOGY (Elective subject)

- CO1: Summarize cell and molecular biology history.
- CO2: Summarize cellular functioning and composition.
- CO3: Describe the chemical foundations of cell biology.
- CO4: Summarize the DNA properties of cell biology.
- CO5: Describe protein structure and function.
- CO6: Describe cellular membrane structure and function.
- CO7: Describe basic molecular genetic mechanisms.
- CO8: Summarize the Cell Cycle

BP809ET. COSMETIC SCIENCE(Theory)

- CO1: Apply basic aspects of cosmetic products
- CO2: Simplify raw materials for cosmetics
- CO3: Appraise toxicological aspects of cosmetics
- CO4: Categorize various cosmetic formulations along with functional evaluation
- CO5: Examine sensorial parameters of cosmetics

BP810 ET. PHARMACOLOGICAL SCREENING METHODS

- CO1: Appreciate the applications of various commonly used laboratory animals.
- CO2: Appreciate and demonstrate the various screening methods used in preclinical research
- CO3: Appreciate and demonstrate the importance of biostatistics and research methodology
- CO4: Design and execute a research hypothesis independently

BP 811 ET. ADVANCED INSTRUMENTATION TECHNIQUES

- CO1: Understand the advanced instruments used and its applications in drug analysis
- CO2: Understand the chromatographic separation and analysis of drugs.
- CO3: Understand the calibration of various analytical instruments

CO4: Know analysis of drugs using various analytical instruments.

BP 812 ET. DIETARY SUPPLEMENTS AND NUTRACEUTICALS

CO1: Understand the need of supplements by the different group of people to maintain healthy life.

CO2: Understand the outcome of deficiencies in dietary supplements.

CO3: Appreciate the components in dietary supplements and the application.

CO4: Appreciate the regulatory and commercial aspects of dietary supplements including health claims.

BACHELOR OF PHARMACY (PRACTICE)

Programme Outcome (POs)

PO1. Assess various disease conditions, read and interpret drug prescription.

PO2. To prepare, package, label and dispense medications, perform non-sterile compounding.

PO3. Understand the professional responsibilities of the pharmacists.

PO4. Appreciate the various therapeutic regimens with their advantages and disadvantages.

PO5. Create awareness in society about the effective and safe use of medicines along with their storage conditions.

PO6. Demonstrate use of medical aids to patients.

PO7. Know the signs and symptoms of the disease.

PO8. Respect patients, healthcare providers and pharmacy co-workers.

PO9. Develop an aptitude for skill development and continuous learning.

PO10. Provide empathy to chronic/emergency/economically back ward patients and their family.

PROGRAM SPECIFIC OUTCOMES:-

PSO1. To develop middle level pharmacy, human resource for good pharmacy practice in the hospital as well as in community settings. To ethically apply their

knowledge and skills of pharmacy with an understanding of realistic constrain for the overall benefit of the society.

- PSO2. To develop close industry-academia ties to generate potential knowledge pool with interpersonal and collaborative skills. To identify the problems and execute the solution in close relation with pharmaceutical industries.
- PSO3. Drug information, patient counselling and awareness and to provide information about the requirement of hospital and community pharmacy and also to develop a sense of team work and awareness among students. To promote rational use of medicine within the health-care facilities and at the community pharmacies.
- PSO4. To produce qualified pharmacist with strong fundamental concept and high technical competence in pharmaceutical science and technology, who are proficient in communicating, understanding the medical languages as it relates to pharmacology, human anatomy and physiology, dispensing medication in varying forms etc.

Course Outcomes:

First-Year

Pathophysiology and Pharmacotherapeutics- I -BPP-101

- CO1. Understand the anatomy and physiology of the respective system Understand the disease process.
- CO2. Know the signs and symptoms of the disease.
- CO3. Appreciate the various therapeutic regimens with their advantages and disadvantages.

Pathophysiology and Pharmacotherapeutics II– BPP-102

- CO1. Understand the anatomy and physiology of respective system
- CO2. Understand the disease process.
- CO3. Know the signs and symptoms of the disease.
- CO4. Appreciate the various therapeutic regimens with their advantages and disadvantages.

Pharmacy Practice- I BPP-103

- CO1. Understand the professional roles of pharmacists in community, hospital and clinical pharmacy areas.
- CO2. Understand the professional responsibilities of the pharmacists.
- CO3. Provide the intended services.

Pharmacy Practice II -BPP-104

- CO1. Understand the professional roles of pharmacists in community, hospital and clinical pharmacy areas.
- CO2. Knowledge about limit tests.
- CO3. Provide the intended services.

Applied Pharmaceutics - BPP-105

- CO1. Understand the formulation principles of various dosage forms
- CO2. Understand the basic principles of stability, storage and administration of various dosage forms
- CO3. Learn about novel drug delivery systems.
- CO4. Understand various pharmacokinetic pathways and optimize the drug therapy.
- CO5. Understand Pro Drugs concept.

Social Pharmacy I - BPP-106

- CO1. Understand the social responsibility of the pharmacists in the society
- CO2. Understand the health policies.
- CO3. Provide health care services to patients.

Second-Year**Pathophysiology and Pharmacotherapeutics III – BPP-201**

- CO1. Understand the anatomy and physiology of the respective system.
- CO2. Understand the disease process.
- CO3. Know the signs and symptoms of the disease.
- CO4. Appreciate the various therapeutic regimens with their advantages and disadvantages

Pathophysiology and Pharmacotherapeutics - IV – BPP-202

- CO1. Understand the anatomy and physiology of the respective system
- CO2. Understand the disease process.
- CO3. Know the signs and symptoms of the disease.
- CO4. Appreciate the various therapeutic regimens with their advantages and disadvantages.

Pharmacy Practice III - BPP-203

- CO1. Understand the professional roles of pharmacists in community, hospital and clinical pharmacy areas.
- CO2. Understand the professional responsibilities of the pharmacists.
- CO3. Provide the intended services.

Pharmacy Practice IV - BPP-204

- CO1. Understand the professional roles of pharmacists in community, hospital and clinical pharmacy areas.
- CO2. Understand the professional responsibilities of the pharmacists.
- CO3. Provide the intended services.

Social Pharmacy - II – BPP-205

- CO1. Understand the social responsibility of the pharmacists in the society.
- CO2. Provide professional services to the patients.

Pharmaceutical Jurisprudence – BPP-206

- CO1. Understand various concepts of the pharmaceutical legislation in India.
- CO2. Know various rules drafted in Drug and Cosmetic Act, Pharmacy Act, NDPS Acts, relevant to pharmacy practice.
- CO3. Know the Consumer Protection Act, PFA Act, DPCO.
- CO4. Understand the labeling requirements and packaging guidelines for drugs and cosmetics.

Master of Pharmacy (M. Pharm)

PROGRAM OUTCOMES (POs)

- PO1 Demonstrate knowledge of the basic pharmaceutical sciences and the ability to acquire, manage and use current information for problem solving.
- PO2 Describe the synthesis, formulation, analysis, pharmacological, pharmacognostical, biotechnological and regulatory aspects of drugs and biopharmaceuticals.
- PO3 Identify the rules and regulations involved in the drug discovery and development, manufacture, distribution and sale of medicines.
- PO4 Develop problem-based learning approach and analytical thinking in his/her academic and professional life.
- PO5 Apply critical thinking skills, including investigation, application, analysis, creativity, evaluation of information, data and documents related to research investigation.
- PO6 Demonstrate the ability to plan and implement professional activities.
- PO7 Act efficiently as a leader in the diverse areas of the profession.

- PO8 Imbibe the skills of scientific communication and research writing.
- PO9 Apply the knowledge and skills gained through education to gain recognition in professional circle and society.
- PO10 Develop interdisciplinary research with other health care communities to provide innovative solutions.
- PO11 Participate in healthcare initiatives to create awareness in society about the effective and safe use of medicines.
- PO12 Demonstrate eco-friendly products and processes to maintain public health.
- PO13 Exercise ethical practices and moral values in personal and professional endeavors.
- PO14 Tackle professional challenges through lifelong learning attitude.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- PSO1 M. Pharm or Master of Pharmacy is a postgraduate Academic Degree which is granted for a course or program in the field of Pharmaceutical Technology.
- PSO2 Pharmaceutics is the discipline of pharmacy that deals with all facets of the process of turning a new chemical entity (NCE) into a medication able to be safely and effectively used by patients in the community.
- PSO3 Pharmaceutics is the science of dosage form design. There are many chemicals with known pharmacological properties but a raw chemical is of no use to a patient
- PSO4 Pharmaceutics deals with the formulation of a pure drug substance into a dosage form. Branches of pharmaceutics include: Pharmacokinetics, Pharmacodynamics, Pharmacoepidemiology, Pharmacogenomics, Pharmacovigilance, Pharmaceutical formulation etc.
- PSO5 Pharmacognosy is an important discipline in the Pharmacy curriculum and deals with the study of crude drugs from natural sources.
- PSO6 The subject has played a diverse role in the discovery, characterization, production and standardization of crude drugs.
- PSO7 In the recent past, the world has witnessed a tremendous renewal of interest in natural products in the area of drug discovery, cosmeceuticals and nutraceuticals. Pharmacognosy now encompasses many fields like botany, ethnobotany, marine biology, microbiology, alternative medicines, chemistry, pharmacology, phytochemistry,

pharmaceutics, clinical pharmacy and molecular biology and has a multidisciplinary approach.

COURSE OUTCOMES (COs)

Pharmaceutical Regulatory Affairs (MRA)

MRA101T. GOOD REGULATORY PRACTICES

- CO-1** The key regulatory and compliance elements with respect to Good Manufacturing Practices, Good Laboratory Practices, Good Automated Laboratory Practices and Good Documentation Practices.
- CO-2** Prepare and implement the check lists and SOPs for various Good Regulatory Practices
- CO-3** Implement Good Regulatory Practices in the Healthcare and related Industries
- CO-4** Prepare for the readiness and conduct of audits and inspections.

MRA102T. Documentation and Regulatory Writing

- CO-1** Know the various documents pertaining to drugs in pharmaceutical industry
- CO-2** Understand the basics of regulatory compilation
- CO-3** Create and assemble the regulation submission as per the requirements of agencies
- CO-4** Follow up the submissions and post approval document requirements

MRA103T. Clinical Research Regulations

- CO-1** History, origin and ethics of clinical and biomedical research and evaluation
- CO-2** Clinical drug, medical device development process and different types and phases of clinical trials
- CO-3** Regulatory requirements and guidance for conduct of clinical trials and research

MRA104T. Regulations and Legislation for Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals in India and Intellectual Property Rights

- CO-1** Know different Acts and guidelines that regulate Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals industry in India.
- CO-2** Understand the approval process and regulatory requirements for Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals

MRA 201T Regulatory Aspects of Drugs & Cosmetics

- CO-1** Process of drug discovery and development and generic product development
- CO-2** Regulatory approval process and registration procedures for API and drug products in US, EU

- CO-3** Cosmetics regulations in regulated and semi-regulated countries
- CO-4** A comparative study of India with other global regulated markets

MRA 202T Regulatory Aspects of Herbal and Biologicals

- CO-1** Know the regulatory Requirements for Biologics and Vaccines
- CO-2** Understand the regulation for newly developed biologics and bio similars
- CO-3** Know the pre-clinical and clinical development considerations of biologics
- CO-4** Understand the Regulatory Requirements of Blood and/or Its Components
Including Blood Products and label requirements

MRA 203T Regulatory Aspects of Medical Devices

- CO-1** Basics of medical devices and IVDs, process of development, ethical and quality considerations
- CO-2** Harmonization initiatives for approval and marketing of medical devices and IVDs
- CO-3** Regulatory approval process for medical devices and IVDs in India, US, Canada, EU, Japan and ASEAN
- CO-4** Clinical evaluation and investigation of medical devices and IVDs

MRA 204T Regulatory Aspects of Food & Nutraceuticals

- CO-1** Know the regulatory Requirements for nutraceuticals
- CO-2** Understand the regulation for registration and labeling of nutraceuticals and food supplements in India, USA and Europe.

MRM301T Research Methodology & Biostatistics

- CO-1** General Research Methodology
- CO-2** Biostatistics
- CO-3** Medical Research
- CO-4** CPCSEA guidelines for laboratory animal facility
- CO-5** Declaration of Helsinki

Pharmaceutics (MPH)

MPH 101T Modern Pharmaceutical Analytical Techniques

- CO-1** Chemicals and Excipients
- CO-2** The analysis of various drugs in single and combination dosage forms
- CO-3** Theoretical and practical skills of the instruments

MPH 102T Drug Delivery Systems

- CO-1** The various approaches for development of novel drug delivery systems.

- CO-2** The criteria for selection of drugs and polymers for the development of delivering system.
- CO-3** The formulation and evaluation of Novel drug delivery systems.

MPH 103T Modern Pharmaceutics

- CO-1** The elements of preformulation studies.
- CO-2** The Active Pharmaceutical Ingredients and Generic drug Product development
- CO-3** Industrial Management and GMP Considerations.
- CO-4** Optimization Techniques & Pilot Plant Scale Up Techniques
- CO-5** Stability Testing, sterilization process & packaging of dosage forms.

MPH 104T Regulatory Affairs

- CO-1** The Concepts of innovator and generic drugs, drug development process
- CO-2** The Regulatory guidance's and guidelines for filing and approval process
- CO-3** Preparation of Dossiers and their submission to regulatory agencies indifferent countries
- CO-4** Post approval regulatory requirements for actives and drug products
- CO-5** Submission of global documents in CTD/ eCTD formats
- CO-6** Clinical trials requirements for approvals for conducting clinical trials
- CO-7** Pharma covigilence and process of monitoring in clinical trials.

MPH 201T Molecular Pharmaceutics (Nano technology & Targeted dds) (NTDS)

- CO-1** The various approaches for development of novel drug delivery systems.
- CO-2** The criteria for selection of drugs and polymers for the development of NTDS
- CO-3** The formulation and evaluation of novel drug delivery systems.

MPH 202T Advanced Biopharmaceutics & Pharmacokinetics

- CO-1** The basic concepts in biopharmaceutics and pharmacokinetics.
- CO-2** The use raw data and derive the pharmacokinetic models and parameters the best describe the process of drug absorption, distribution, metabolism and elimination.
- CO-3** The critical evaluation of biopharmaceutic studies involving drug product equivalency.
- CO-4** The design and evaluation of dosage regimens of the drugs using pharmacokinetic and biopharmaceutic parameters.
- CO-5** The potential clinical pharmacokinetic problems and application of basics of pharmacokinetic

MPH 203T Computer Aided Drug Development

- CO-1** History of Computers in Pharmaceutical Research and Development
- CO-2** Computational Modeling of Drug Disposition

- CO-3** Computers in Preclinical Development
- CO-4** Optimization Techniques in Pharmaceutical Formulation
- CO-5** Computers in Market Analysis
- CO-6** Computers in Clinical Development
- CO-7** Artificial Intelligence (AI) and Robotics
- CO-8** Computational fluid dynamics (CFD)

MPH 204T Cosmetics and Cosmeceuticals

- CO-1** Key ingredients used in cosmetics and cosmeceuticals.
- CO-2** Key building blocks for various formulations.
- CO-3** Current technologies in the market
- CO-4** Various key ingredients and basic science to develop cosmetics and cosmeceutical
- CO-5** Scientific knowledge to develop cosmetics and cosmeceuticals with desired Safety, stability, and efficacy.

MRM301T Research Methodology & Biostatistics

After completion of course learner will be able to understand the following

- CO-1 General Research Methodology
- CO-2 Biostatistics
- CO-3 Medical Research
- CO-4 CPCSEA guidelines for laboratory animal facility
- CO-5 Declaration of Helsinki

PHARMACOLOGY (MPL)

MPL 101T Modern Pharmaceutical Analytical Techniques

- CO-1** Chemicals and Excipients
- CO-2** The analysis of various drugs in single and combination dosage forms
- CO-3** Theoretical and practical skills of the instruments

MPL 102T Advanced Pharmacology – I

- CO-1 Discuss the pathophysiology and pharmacotherapy of certain diseases
- CO-2 Explain the mechanism of drug actions at cellular and molecular level
- CO-3 Understand the adverse effects, contraindications and clinical uses of drugs used in treatment of diseases

MPL 103T Pharmacological and Toxicological Screening Methods – I

- CO-1 Appraise the regulations and ethical requirement for the usage of experimental animals.

- CO-2 Describe the various animals used in the drug discovery process and good laboratory practices in maintenance and handling of experimental animals
- CO-3 Describe the various newer screening methods involved in the drug discovery process

Appreciate and correlate the preclinical data to humans

MPL 104T Cellular and Molecular Pharmacology

- CO-1 Explain the receptor signal transduction processes.
- CO-2 Explain the molecular pathways affected by drugs.
- CO-3 Appreciate the applicability of molecular pharmacology and biomarkers in drug discovery process.
- CO-4 Demonstrate molecular biology techniques as applicable for pharmacology

MPL 201T Advanced Pharmacology – II

- CO-1 Explain the mechanism of drug actions at cellular and molecular level
- CO-2 Discuss the Pathophysiology and pharmacotherapy of certain diseases
- CO-3 Understand the adverse effects, contraindications and clinical uses of drugs used in treatment of diseases

MPL 202T Pharmacological and Toxicological Screening Methods-II

- CO-1 Explain the various types of toxicity studies.
- CO-2 Appreciate the importance of ethical and regulatory requirements for toxicity studies.
- CO-3 Demonstrate the practical skills required to conduct the preclinical toxicity studies.

MPL 203T Principles of Drug Discovery

- CO-1 Explain the various stages of drug discovery.
- CO-2 Appreciate the importance of the role of genomics, proteomics and bioinformatics in drug discovery
- CO-3 Explain various targets for drug discovery.
- CO-4 Explain various lead seeking method and lead optimization
- CO-5 Appreciate the importance of the role of computer aided drug design in drug discovery

MPL 204T Clinical Research and Pharmacovigilance

- CO-1 Explain the regulatory requirements for conducting clinical trial
- CO-2 Demonstrate the types of clinical trial designs
- CO-3 Explain the responsibilities of key players involved in clinical trials
- CO-4 Execute safety monitoring, reporting and close-out activities
- CO-5 Explain the principles of Pharmacovigilance
- CO-6 Detect new adverse drug reactions and their assessment

CO-7 Perform the adverse drug reaction reporting systems and communication in Pharmacovigilance

MRM301T Research Methodology & Biostatistics

After completion of course learner will able to understand the following

CO-1 General Research Methodology

CO-2 Biostatistics

CO-3 Medical Research

CO-4 CPCSEA guidelines for laboratory animal facility

CO-5 Declaration of Helsinki

PHARMACOGNOSY (MPG)

MPG 101T Modern Pharmaceutical Analytical Techniques

CO-1 The analysis of various drugs in single and combination dosage forms

CO-2 Theoretical and practical skills of the instruments

MPG 102T Advanced Pharmacognosy – I

CO-1 Advances in the cultivation and production of drugs

CO-2 Various phyto-pharmaceuticals and their source, its utilization and medicinal value.

CO-3 Various nutraceuticals/herbs and their health benefits

CO-4 Drugs of marine origin

CO-5 Pharmacovigilance of drugs of natural origin

MPG 103T Phytochemistry

CO-1 Different classes of phytoconstituents, their biosynthetic pathways, their properties, extraction and general process of natural product drug discovery

CO-2 Phytochemical fingerprinting and structure elucidation of phytoconstituents.

MPG 104T Industrial Pharmacognostical Technology

CO-1 The requirements for setting up the herbal/natural drug industry.

CO-2 The guidelines for quality of herbal/natural medicines and regulatory issues.

CO-3 The patenting/IPR of herbals/natural drugs and trade of raw and finished materials.

MPG 201T Medicinal Plant Biotechnology

CO-1 Know the process like genetic engineering in medicinal plants for higher yield of Phytopharmaceuticals.

CO-2 Use the biotechnological techniques for obtaining and improving the quality of natural products/medicinal plants

MPG 202T Advanced Pharmacognosy – II

CO-1 Validation of herbal remedies

CO-2 Methods of detection of adulteration and evaluation techniques for the herbal drugs

CO-3 Methods of screening of herbals for various biological properties

MPG 203T Indian Systems of Medicine

CO-1 To understand the basic principles of various Indian systems of medicine

CO-2 To know the clinical research of traditional medicines, Current Good Manufacturing Practice of Indian systems of medicine and their formulations.

MPG 204T Herbal Cosmetics

CO-1 Understand the basic principles of various herbal/natural cosmetic preparations

CO-2 Current Good Manufacturing Practices of herbal/natural cosmetics as per the regulatory authorities

MRM301T Research Methodology & Biostatistics

After completion of course learner will be able to understand the following

CO-1 General Research Methodology

CO-2 Biostatistics

CO-3 Medical Research

CO-4 CPCSEA guidelines for laboratory animal facility

CO-5 Declaration of Helsinki

Master of Pharmacy (M. Pharm)

Pharmaceutical Regulatory Affairs (MRA)

COURSE OUTCOMES (COs)

MRA101T. GOOD REGULATORY PRACTICES

CO-1 The key regulatory and compliance elements with respect to Good Manufacturing Practices, Good Laboratory Practices, Good Automated Laboratory Practices and Good Documentation Practices.

CO-2 Prepare and implement the check lists and SOPs for various Good Regulatory Practices

CO-3 Implement Good Regulatory Practices in the Healthcare and related Industries

CO-4 Prepare for the readiness and conduct of audits and inspections.

MRA102T. Documentation and Regulatory Writing

CO-1 Know the various documents pertaining to drugs in pharmaceutical industry

CO-2 Understand the basics of regulatory compilation

CO-3 Create and assemble the regulation submission as per the requirements of agencies

CO-4 Follow up the submissions and post approval document requirements

MRA103T. Clinical Research Regulations

CO-1 History, origin and ethics of clinical and biomedical research and evaluation

CO-2 Clinical drug, medical device development process and different types and phases of clinical trials

CO-3 Regulatory requirements and guidance for conduct of clinical trials and research

MRA104T. Regulations and Legislation for Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals in India and Intellectual Property Rights

CO-1 Know different Acts and guidelines that regulate Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals industry in India.

CO-2 Understand the approval process and regulatory requirements for Drugs & Cosmetics, Medical Devices, Biologicals & Herbals, and Food & Nutraceuticals

MRA 201T Regulatory Aspects of Drugs & Cosmetics

CO-1 Process of drug discovery and development and generic product development

CO-2 Regulatory approval process and registration procedures for API and drug products in US, EU

CO-3 Cosmetics regulations in regulated and semi-regulated countries

CO-4 A comparative study of India with other global regulated markets

MRA 202T Regulatory Aspects of Herbal and Biologicals

CO-1 Know the regulatory Requirements for Biologics and Vaccines

CO-2 Understand the regulation for newly developed biologics and biosimilars

CO-3 Know the pre-clinical and clinical development considerations of biologics

CO-4 Understand the Regulatory Requirements of Blood and/or Its Components Including Blood Products and label requirements

MRA 203T Regulatory Aspects of Medical Devices

CO-1 Basics of medical devices and IVDs, process of development, ethical and quality considerations

CO-2 Harmonization initiatives for approval and marketing of medical devices and IVDs

CO-3 Regulatory approval process for medical devices and IVDs in India, US, Canada, EU, Japan and ASEAN

CO-4 Clinical evaluation and investigation of medical devices and IVDs

MRA 204T Regulatory Aspects of Food & Nutraceuticals

CO-1 Know the regulatory Requirements for nutraceuticals

CO-2 Understand the regulation for registration and labeling of nutraceuticals and food supplements in India, USA and Europe.

MRM301T Research Methodology & Biostatistics

After completion of course learner will be able to understand the following

CO-1 General Research Methodology

CO-2 Biostatistics

CO-3 Medical Research

CO-4 CPCSEA guidelines for laboratory animal facility

CO-5 Declaration of Helsinki

Master of Pharmacy (M. Pharm)

Pharmaceutics (MPH)

COURSE OUTCOMES (COs)

MPH 101T Modern Pharmaceutical Analytical Techniques

CO-1 Chemicals and Excipients

CO-2 The analysis of various drugs in single and combination dosage forms

CO-3 Theoretical and practical skills of the instruments

MPH 102T Drug Delivery Systems

CO-1 The various approaches for development of novel drug delivery systems.

CO-2 The criteria for selection of drugs and polymers for the development of delivering system.

CO-3 The formulation and evaluation of Novel drug delivery systems..

MPH 103T Modern Pharmaceutics

CO-1 The elements of preformulation studies.

CO-2 The Active Pharmaceutical Ingredients and Generic drug Product development

CO-3 Industrial Management and GMP Considerations.

CO-4 Optimization Techniques & Pilot Plant Scale Up Techniques

CO-5 Stability Testing, sterilization process & packaging of dosage forms.

MPH 104T Regulatory Affairs

CO-1 The Concepts of innovator and generic drugs, drug development process

CO-2 The Regulatory guidance's and guidelines for filing and approval process

CO-3 Preparation of Dossiers and their submission to regulatory agencies in different countries

- CO-4 Post approval regulatory requirements for actives and drug products
- CO-5 Submission of global documents in CTD/ eCTD formats
- CO-6 Clinical trials requirements for approvals for conducting clinical trials
- CO-7 Pharmacovigilance and process of monitoring in clinical trials.

MPH 201T Molecular Pharmaceutics (nano technology & Targeted dds) (NTDS)

- CO-1 The various approaches for development of novel drug delivery systems.
- CO-2 The criteria for selection of drugs and polymers for the development of NTDS
- CO-3 The formulation and evaluation of novel drug delivery systems.

MPH 202T Advanced Biopharmaceutics & Pharmacokinetics

- CO-1 The basic concepts in biopharmaceutics and pharmacokinetics.
- CO-2 The use raw data and derive the pharmacokinetic models and parameters the best describe the process of drug absorption, distribution, metabolism and elimination.
- CO-3 The critical evaluation of biopharmaceutic studies involving drug product equivalency.
- CO-4 The design and evaluation of dosage regimens of the drugs using pharmacokinetic and biopharmaceutic parameters.
- CO-5 The potential clinical pharmacokinetic problems and application of basics of pharmacokinetic

MPH 203T Computer Aided Drug Development

- CO-1 History of Computers in Pharmaceutical Research and Development
- CO-2 Computational Modeling of Drug Disposition
- CO-3 Computers in Preclinical Development
- CO-4 Optimization Techniques in Pharmaceutical Formulation
- CO-5 Computers in Market Analysis
- CO-6 Computers in Clinical Development
- CO-7 Artificial Intelligence (AI) and Robotics
- CO-8 Computational fluid dynamics (CFD)

MPH 204T Cosmetics and Cosmeceuticals

- CO-1 Key ingredients used in cosmetics and cosmeceuticals.
- CO-2 Key building blocks for various formulations.
- CO-3 Current technologies in the market
- CO-4 Various key ingredients and basic science to develop cosmetics and cosmeceutical
- CO-5 Scientific knowledge to develop cosmetics and cosmeceuticals with desired Safety, stability, and efficacy.

MRM301T Research Methodology & Biostatistics

- CO-1 General Research Methodology
- CO-2 Biostatistics
- CO-3 Medical Research

CO-4 CPCSEA guidelines for laboratory animal facility
CO-5 Declaration of Helsinki

Master of Pharmacy (M. Pharm)

PHARMACOLOGY (MPL)

COURSE OUTCOMES (COs)

MPL 101T Modern Pharmaceutical Analytical Techniques

- CO-1** Chemicals and Excipients
- CO-2** The analysis of various drugs in single and combination dosage forms
- CO-3** Theoretical and practical skills of the instruments

MPL 102T Advanced Pharmacology – I

- CO-1** Discuss the pathophysiology and pharmacotherapy of certain diseases
- CO-2** Explain the mechanism of drug actions at cellular and molecular level
- CO-3** Understand the adverse effects, contraindications and clinical uses of drugs used in treatment of diseases

MPL 103T Pharmacological and Toxicological Screening Methods – I

- CO-1** Appraise the regulations and ethical requirement for the usage of experimental animals.
- CO-2** Describe the various animals used in the drug discovery process and good laboratory practices in maintenance and handling of experimental animals
- CO-3** Describe the various newer screening methods involved in the drug discovery process

MPL 104T Cellular and Molecular Pharmacology

- CO-1** Explain the receptor signal transduction processes.
- CO-2** Explain the molecular pathways affected by drugs.
- CO-3** Appreciate the applicability of molecular pharmacology and biomarkers in drug discovery process.
- CO-4** Demonstrate molecular biology techniques as applicable for pharmacology

MPL 201T Advanced Pharmacology – II

- CO-1** Explain the mechanism of drug actions at cellular and molecular level
- CO-2** Discuss the Pathophysiology and pharmacotherapy of certain diseases
- CO-3** Understand the adverse effects, contraindications and clinical uses of drugs used in treatment of diseases

MPL 202T Pharmacological and Toxicological Screening Methods-II

- CO-1** Explain the various types of toxicity studies.

- CO-2 Appreciate the importance of ethical and regulatory requirements for toxicity studies.
- CO-3 Demonstrate the practical skills required to conduct the preclinical toxicity studies.

MPL 203T Principles of Drug Discovery

- CO-1 Explain the various stages of drug discovery.
- CO-2 Appreciate the importance of the role of genomics, proteomics and bioinformatics in drug discovery
- CO-3 Explain various targets for drug discovery.
- CO-4 Explain various lead seeking method and lead optimization
- CO-5 Appreciate the importance of the role of computer aided drug design in drug discovery

MPL 204T Clinical Research and Pharmacovigilance

- CO-1 Explain the regulatory requirements for conducting clinical trial
- CO-2 Demonstrate the types of clinical trial designs
- CO-3 Explain the responsibilities of key players involved in clinical trials
- CO-4 Execute safety monitoring, reporting and close-out activities
- CO-5 Explain the principles of Pharmacovigilance
- CO-6 Detect new adverse drug reactions and their assessment
- CO-7 Perform the adverse drug reaction reporting systems and communication in Pharmacovigilance

MRM301T Research Methodology & Biostatistics

After completion of course learner will be able to understand the following

- CO-1 General Research Methodology
- CO-2 Biostatistics
- CO-3 Medical Research
- CO-4 CPCSEA guidelines for laboratory animal facility
- CO-5 Declaration of Helsinki

Master of Pharmacy (M. Pharm)

PHARMACOGNOSY (MPG)

COURSE OUTCOMES (COs)

MPG 101T Modern Pharmaceutical Analytical Techniques

- CO-1 The analysis of various drugs in single and combination dosage forms

CO-2 Theoretical and practical skills of the instruments

MPG 102T Advanced Pharmacognosy – I

CO-1 Advances in the cultivation and production of drugs

CO-2 Various phyto-pharmaceuticals and their source, its utilization and medicinal value.

CO-3 Various nutraceuticals/herbs and their health benefits

CO-4 Drugs of marine origin

CO-5 Pharmacovigilance of drugs of natural origin

MPG 103T Phytochemistry

CO-1 Different classes of phytoconstituents, their biosynthetic pathways, their properties, extraction and general process of natural product drug discovery

CO-2 Phytochemical fingerprinting and structure elucidation of phytoconstituents.

MPG 104T Industrial Pharmacognostical Technology

CO-1 The requirements for setting up the herbal/natural drug industry.

CO-2 The guidelines for quality of herbal/natural medicines and regulatory issues.

CO-3 The patenting/IPR of herbals/natural drugs and trade of raw and finished materials.

MPG 201T Medicinal Plant Biotechnology

CO-1 Know the process like genetic engineering in medicinal plants for higher yield of Phytopharmaceuticals.

CO-2 Use the biotechnological techniques for obtaining and improving the quality of natural products/medicinal plants

MPG 202T Advanced Pharmacognosy – II

CO-1 Validation of herbal remedies

CO-2 Methods of detection of adulteration and evaluation techniques for the herbal drugs

CO-3 Methods of screening of herbals for various biological properties

MPG 203T Indian Systems of Medicine

CO-1 To understand the basic principles of various Indian systems of medicine

CO-2 To know the clinical research of traditional medicines, Current Good Manufacturing Practice of Indian systems of medicine and their formulations.

MPG 204T Herbal Cosmetics

CO-1 Understand the basic principles of various herbal/natural cosmetic preparations

CO-2 Current Good Manufacturing Practices of herbal/natural cosmetics as per the regulatory authorities

MRM301T Research Methodology & Biostatistics

After completion of course learner will be able to understand the following

CO-1 General Research Methodology

CO-2 Biostatistics

CO-3 Medical Research

CO-4 CPCSEA guidelines for laboratory animal facility

CO-5 Declaration of Helsinki